SfM-FailSafe

Failure prediction and adaptive track weighting for sparse 3D reconstruction

Generated 2026-11-10 00:21 | Python 3.12.10 | COLMAP-based pipeline

Real COLMAP runs	9
Synthetic validation runs	16
Report artifacts	REPORT.pdf

Executive summary

Clean COLMAP baseline registered 38/38 cameras with 20445 3D points and mean reprojection error 0.7544 px. Real runs: 5 success / 4 failure.

Post-mapper refinement: baseline 0.7544 px; best variant uniform 0.7454 px. Uniform BA is the control; weighted gains are not assumed.

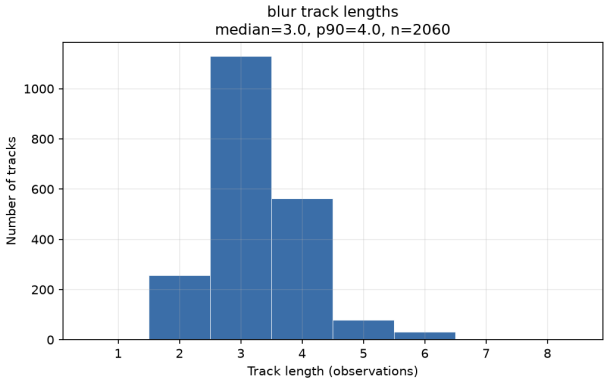
Real run matrix

run	cond	status	images	registered	ratio	points	mean_px	median_px	track_med	label	sec
blur	blur	success	12	9	0.750	2060	0.627	0.539	3.000	success	17.834
blur_mild	blur_mild	success	12	9	0.750	2031	0.637	0.547	3.000	success	12.999
blur_severe	blur_severe	success	12	9	0.750	3284	0.547	0.424	3.000	success	26.649
clean	clean	success	38	38	1.000	20445	0.754	0.643	4.000	success	92.971
degraded_matches	degraded_matches	success	38	20	0.526	2836	0.713	0.620	3.000	failure	152.011
degraded_matches_severe	degraded_matches_severe	success	38	2	0.053	128	0.386	0.232	2.000	failure	7.915
limited_views	limited_views	success	8	6	0.750	2270	0.382	0.273	2.000	success	13.046
limited_views_severe	limited_views_severe	failed	4	0	0.000	0	n/a	n/a	n/a	failure	2.000
smoke	smoke	success	6	3	0.500	544	0.128	0.073	2.000	failure	4.308

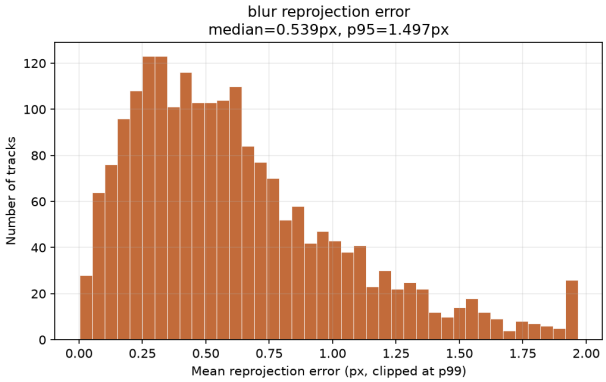
Per-run analysis and plots

Run: blur

Condition	blur
Data kind	real
Status	success
Registered	9/12
Registration ratio	0.750
3D points	2060
Mean / median reprojection (px)	0.6269 / 0.5391
Track length mean / median / p90	3.28 / 3.00 / 4.00
Success label	success
Failure reasons	-
Runtime (s)	17.8



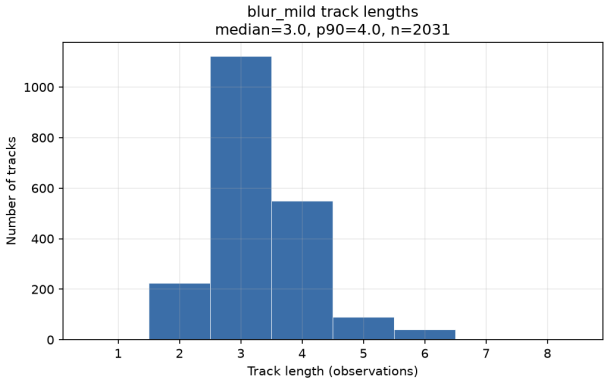
Track length distribution



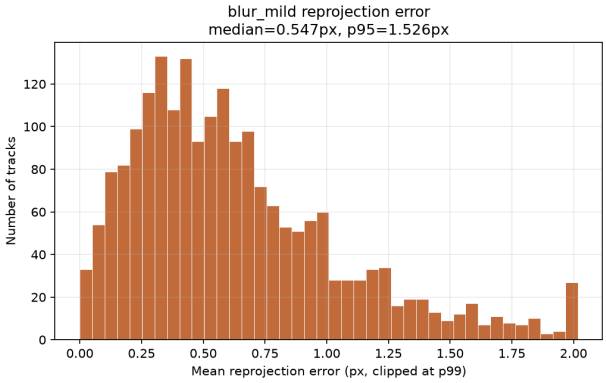
Track mean reprojection error

Run: blur_mild

Condition	blur_mild
Data kind	real
Status	success
Registered	9/12
Registration ratio	0.750
3D points	2031
Mean / median reprojection (px)	0.6370 / 0.5466
Track length mean / median / p90	3.32 / 3.00 / 4.00
Success label	success
Failure reasons	-
Runtime (s)	13.0



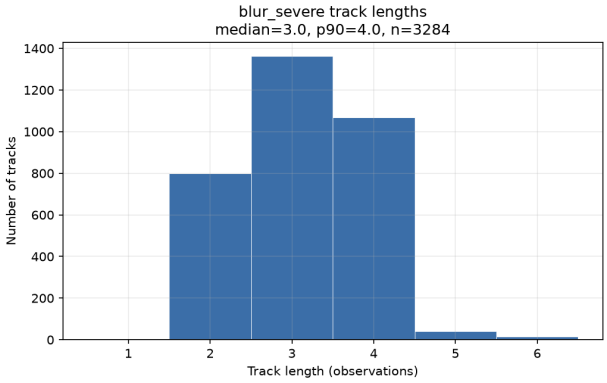
Track length distribution



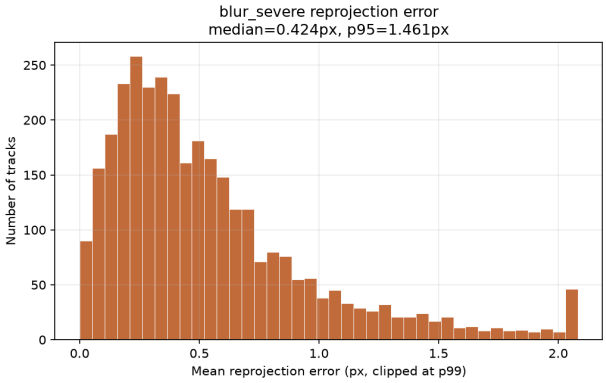
Track mean reprojection error

Run: blur_severe

Condition	blur_severe
Data kind	real
Status	success
Registered	9/12
Registration ratio	0.750
3D points	3284
Mean / median reprojection (px)	0.5470 / 0.4237
Track length mean / median / p90	3.12 / 3.00 / 4.00
Success label	success
Failure reasons	-
Runtime (s)	26.6



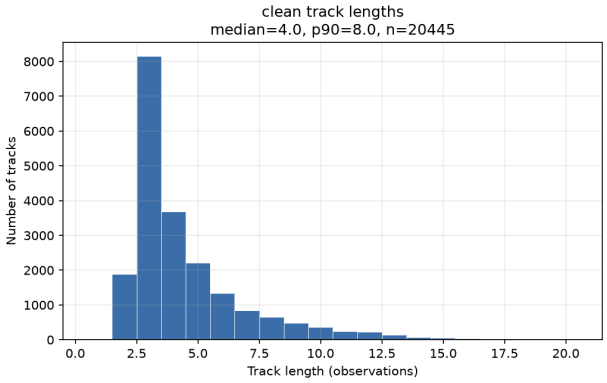
Track length distribution



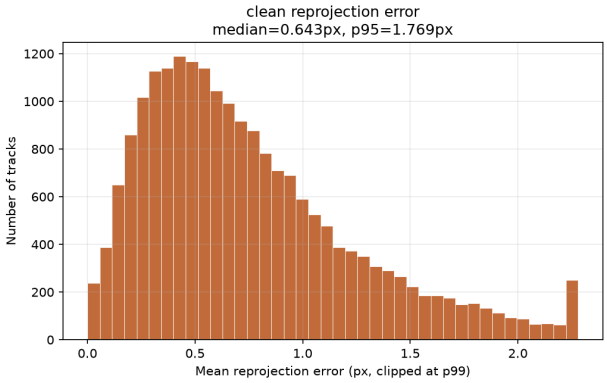
Track mean reprojection error

Run: clean

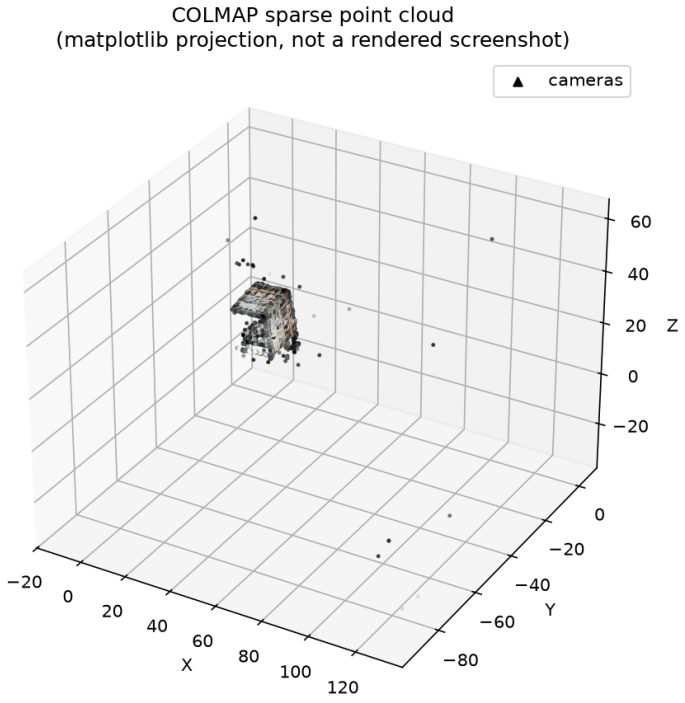
Condition	clean
Data kind	real
Status	success
Registered	38/38
Registration ratio	1.000
3D points	20445
Mean / median reprojection (px)	0.7544 / 0.6431
Track length mean / median / p90	4.48 / 4.00 / 8.00
Success label	success
Failure reasons	-
Runtime (s)	93.0



Track length distribution



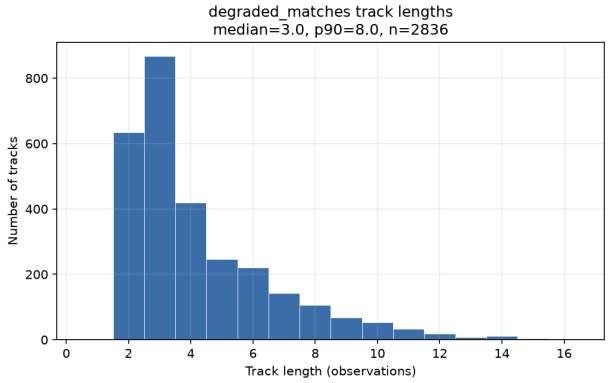
Track mean reprojection error



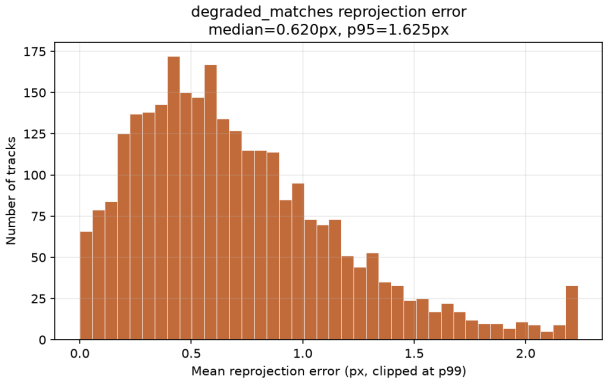
Sparse point cloud projection (matplotlib, not a rendered screenshot)

Run: degraded_matches

Condition	degraded_matches
Data kind	real
Status	success
Registered	20/38
Registration ratio	0.526
3D points	2836
Mean / median reprojection (px)	0.7130 / 0.6198
Track length mean / median / p90	4.25 / 3.00 / 8.00
Success label	failure
Failure reasons	registration_ratio 0.526 < 0.600
Runtime (s)	152.0



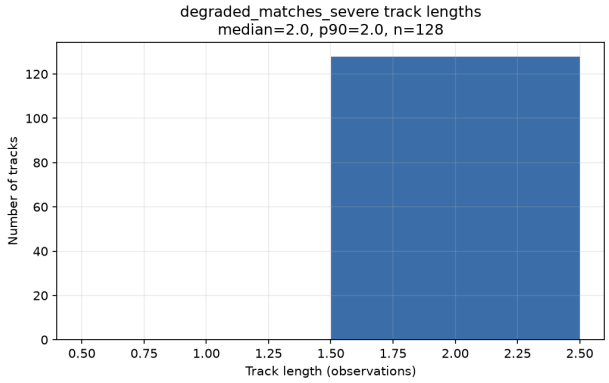
Track length distribution



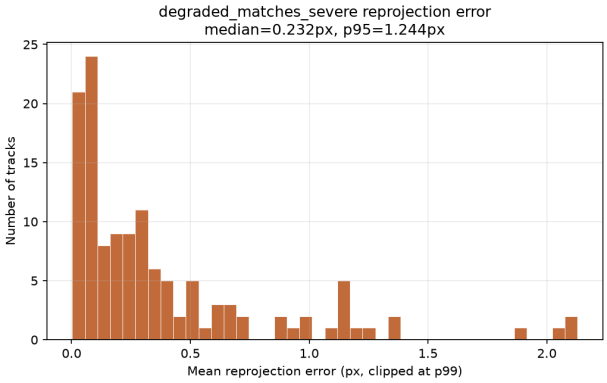
Track mean reprojection error

Run: degraded_matches_severe

Condition	degraded_matches_severe
Data kind	real
Status	success
Registered	2/38
Registration ratio	0.053
3D points	128
Mean / median reprojection (px)	0.3858 / 0.2319
Track length mean / median / p90	2.00 / 2.00 / 2.00
Success label	failure
Failure reasons	registration_ratio 0.053 < 0.600; num_points3D 128 < 1000
Runtime (s)	7.9



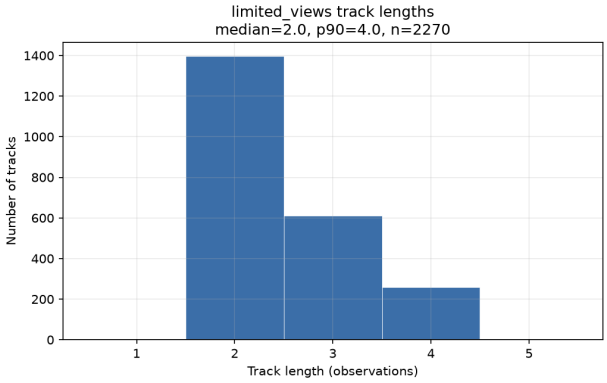
Track length distribution



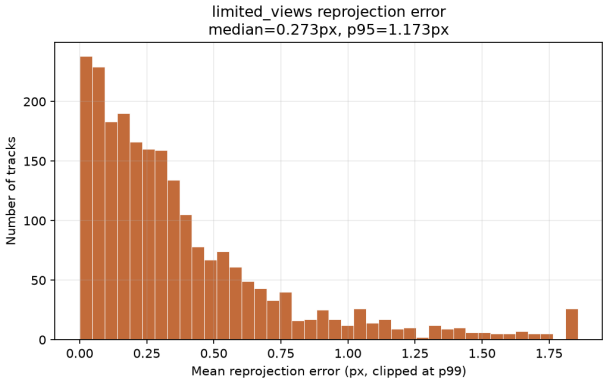
Track mean reprojection error

Run: limited_views

Condition	limited_views
Data kind	real
Status	success
Registered	6/8
Registration ratio	0.750
3D points	2270
Mean / median reprojection (px)	0.3820 / 0.2726
Track length mean / median / p90	2.50 / 2.00 / 4.00
Success label	success
Failure reasons	-
Runtime (s)	13.0



Track length distribution



Track mean reprojection error

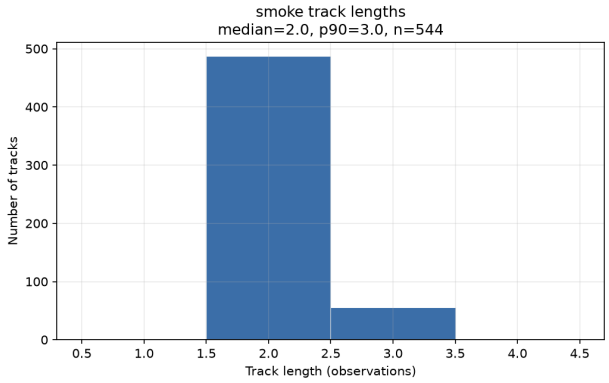
Run: limited_views_severe

Condition	limited_views_severe
Data kind	real
Status	failed
Registered	0/4
Registration ratio	0.000
3D points	0
Mean / median reprojection (px)	n/a / n/a
Track length mean / median / p90	n/a / n/a / n/a
Success label	failure
Failure reasons	mapper_failed; registration_ratio 0.000 < 0.600; num_points3D 0 < 1000
Runtime (s)	2.0

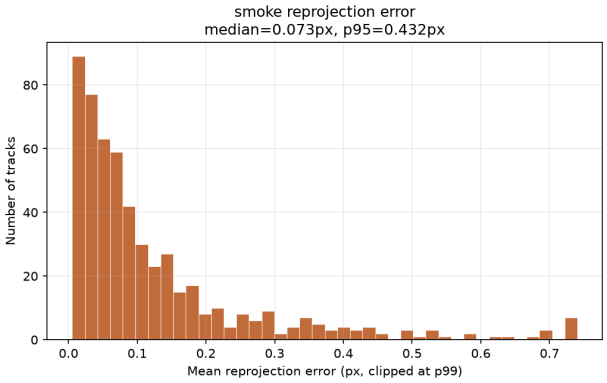
No analysis plots for this run.

Run: smoke

Condition	smoke
Data kind	real
Status	success
Registered	3/6
Registration ratio	0.500
3D points	544
Mean / median reprojection (px)	0.1277 / 0.0735
Track length mean / median / p90	2.11 / 2.00 / 3.00
Success label	failure
Failure reasons	registration_ratio 0.500 < 0.600; num_points3D 544 < 1000
Runtime (s)	4.3



Track length distribution



Track mean reprojection error

Failure prediction

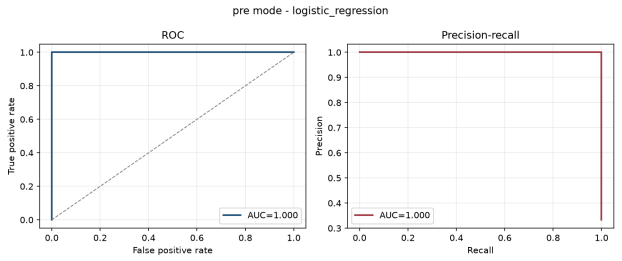
Pre-reconstruction model (leakage-free)

Mode	pre
Leakage-free	True
Runs	9
Class counts	{'0': 5, '1': 4}
Features	38
Split	stratified_holdout
Best model	logistic_regression
Threshold	0.5000
Preliminary	True

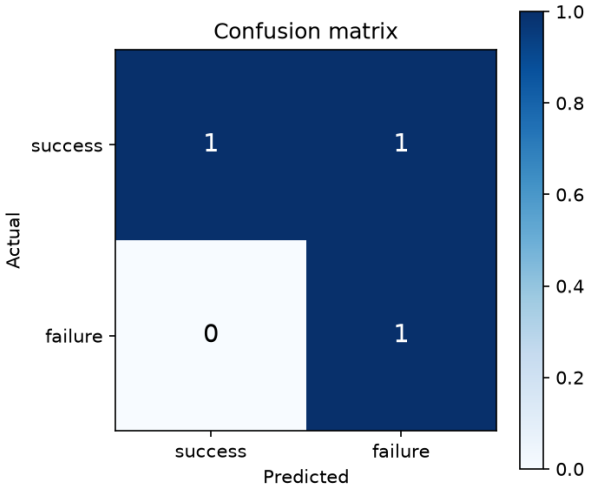
model	accuracy	precision	recall	f1	roc_auc	avg_prec	tp	tn	fp	fn
logistic_regression	0.667	0.500	1.000	0.667	1.000	1.000	1	1	1	0
random_forest	0.333	0.000	0.000	0.000	0.500	0.500	0	1	1	1
gradient_boosting	0.000	0.000	0.000	0.000	0.000	0.333	0	0	2	1

Held-out runs: limited_views, smoke, clean
Leave-one-condition-out (random forest):

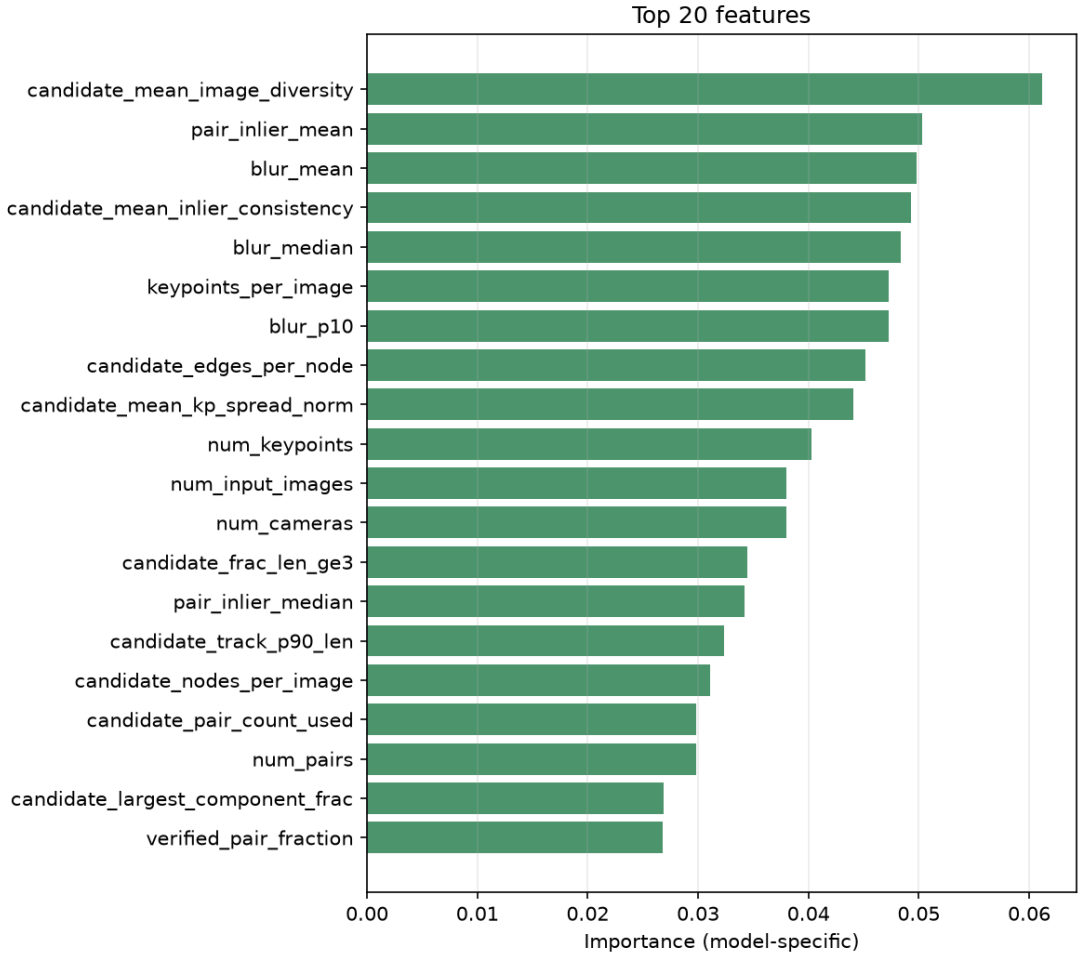
n	threshold	accuracy	precision	recall	f1	specificity	roc_auc	avg_prec	tp	tn	fp	fn	pred_pos	actual_pos
9.000	0.500	0.667	0.667	0.500	0.571	0.800	0.750	0.747	2.000	4.000	1.000	2.000	3.000	4.000



ROC / precision-recall



Confusion matrix



Feature importance

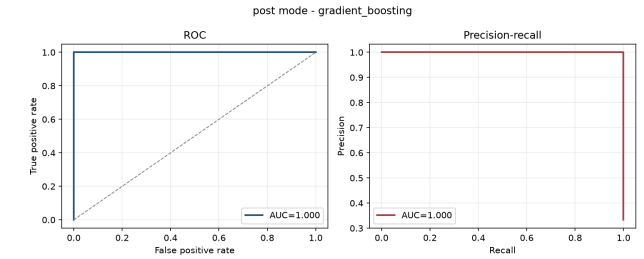
Post-reconstruction diagnostic model (NOT leakage-free)

Mode	post
Leakage-free	False
Runs	9
Class counts	{'0': 5, '1': 4}
Features	23
Split	stratified_holdout
Best model	gradient_boosting
Threshold	0.5000
Preliminary	True

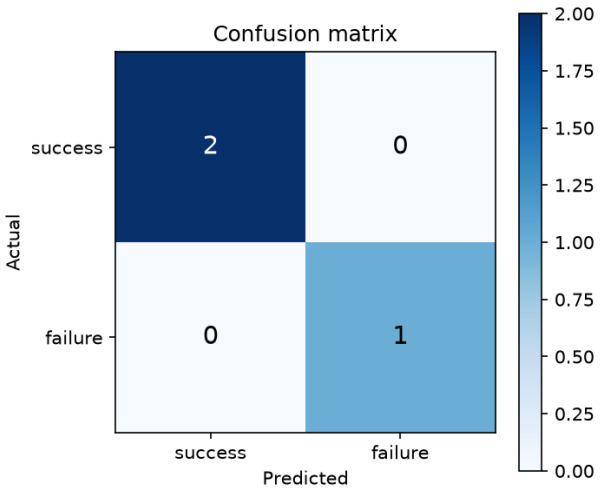
model	accuracy	precision	recall	f1	roc_auc	avg_prec	tp	tn	fp	fn
logistic_regression	0.667	0.000	0.000	0.000	0.000	0.333	0	2	0	1
random_forest	0.333	0.000	0.000	0.000	0.500	0.500	0	1	1	1
gradient_boosting	1.000	1.000	1.000	1.000	1.000	1.000	1	2	0	0

Held-out runs: limited_views, smoke, clean
Leave-one-condition-out (random forest):

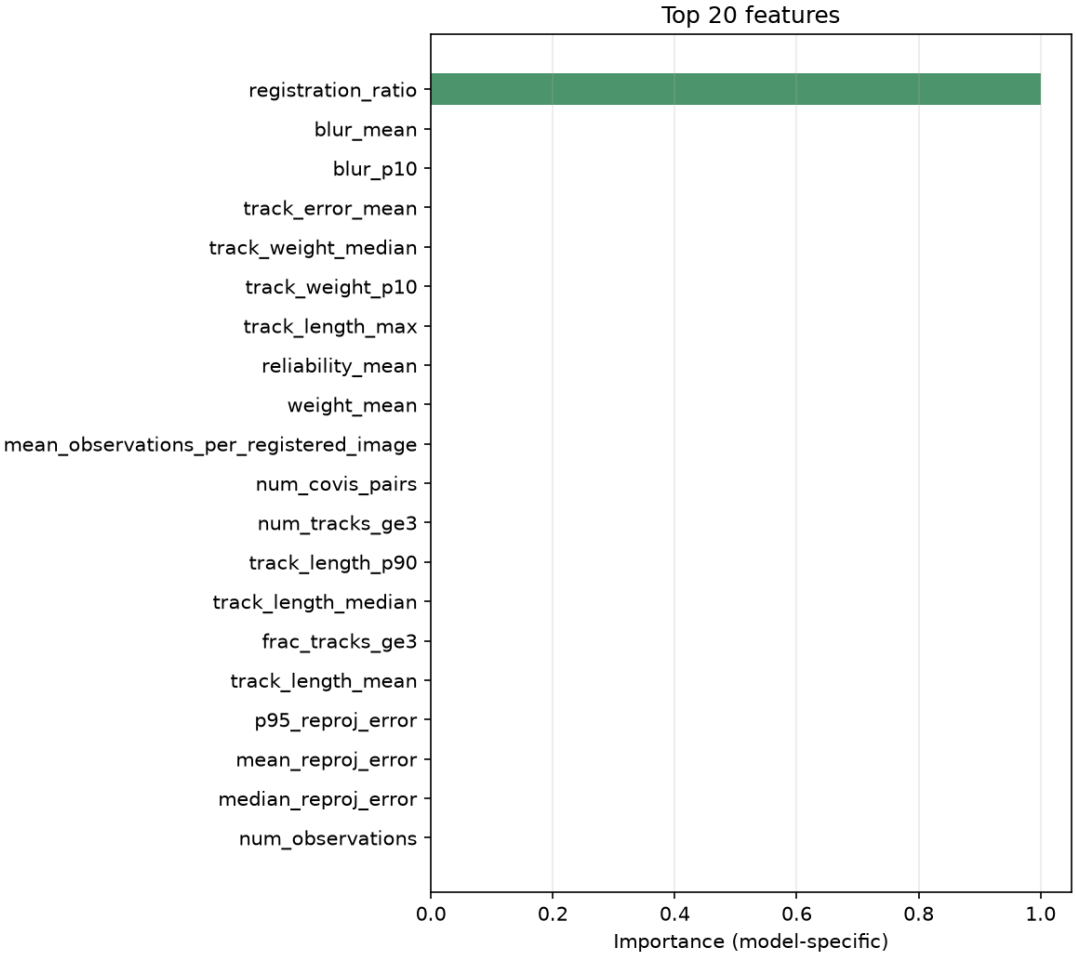
n	threshold	accuracy	precision	recall	f1	specificity	roc_auc	avg_prec	tp	tn	fp	fn	pred_pos	actual_pos
9.000	0.500	0.556	0.500	0.250	0.333	0.800	0.650	0.715	1.000	4.000	1.000	3.000	2.000	4.000



ROC / precision-recall



Confusion matrix



Feature importance

Baseline vs adaptive track weighting

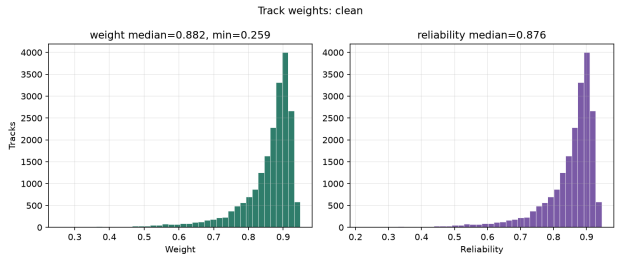
variant	engine	points	mean_err	med_err	trk_med	w_mean	success_label	sec	d_mean
baseline	colmap_baseline	20445	0.754	0.643	4.000	0.854	0	92.971	0.000
uniform	scipy_least_squares	20445	0.745	0.637	4.000	1.000	0	39.034	-0.009
length	scipy_least_squares	20445	0.747	0.638	4.000	0.843	0	31.008	-0.008
full	scipy_least_squares	20445	0.747	0.637	4.000	0.854	0	38.279	-0.008

Weight distributions

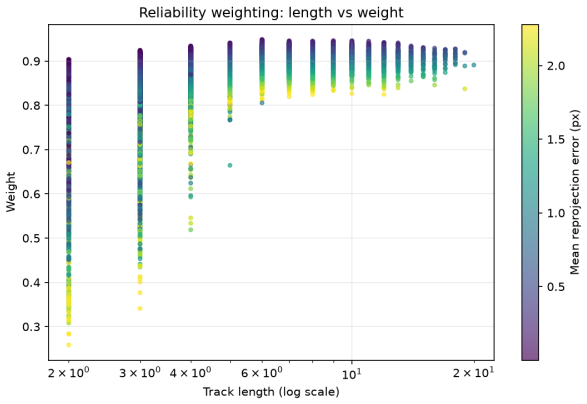
mode	count	w_mean	weight_std	weight_min	weight_p10	weight_median	weight_max	rel_mean	w<0.1	w>0.9
uniform	20445	1.000	0.000	1.000	1.000	1.000	1.000	1.000	0.000	1.000
length	20445	0.843	0.098	0.616	0.792	0.883	0.955	0.835	0.000	0.329
full	20445	0.854	0.083	0.259	0.757	0.882	0.949	0.847	0.000	0.319

Refinement diagnostics

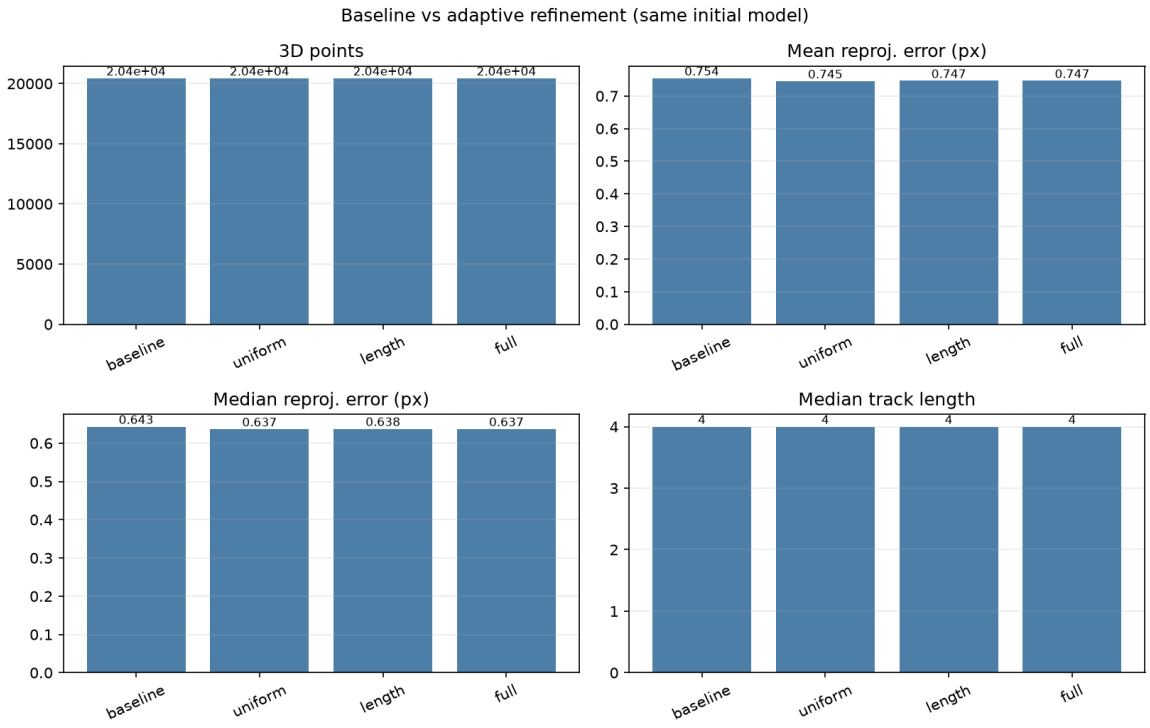
mode	status	engine	sel_pts	obs	rmse0	rmse1	retri	runtime_s
uniform	ok	scipy_least_squares	20445	91619	1.004	1.007	n/a	39.034
length	ok	scipy_least_squares	20445	91619	1.004	1.007	n/a	31.008
full	ok	scipy_least_squares	20445	91619	1.004	1.006	n/a	38.279



Track weight distribution



Track length vs weight



Baseline vs refinement metrics (same initial model)

Synthetic pipeline validation

WARNING: generated sphere scene with a simulated mapper. Not COLMAP/ETH3D results; validation only.

run_name	cond	status	label_name	imgs	reg	points	mean_err
clean_s0	clean	success	success	12	12	2672	0.660
clean_s1	clean	success	success	12	12	2660	0.647
blur_mild_s0	blur_mild	success	success	12	12	2672	1.155
blur_mild_s1	blur_mild	success	success	12	12	2660	1.132
blur_severe_s0	blur_severe	success	failure	12	12	2672	2.309
blur_severe_s1	blur_severe	success	failure	12	12	2660	2.265
limited_views_s0	limited_views	success	success	6	6	2439	0.990
limited_views_s1	limited_views	success	success	6	6	2452	0.964
limited_views_severe_s0	limited_views_severe	success	failure	4	2	302	1.682
limited_views_severe_s1	limited_views_severe	success	success	4	4	1200	1.651
degraded_matches_mild_s0	degraded_matches_mild	success	success	12	12	2672	1.155
degraded_matches_mild_s1	degraded_matches_mild	success	success	12	12	2660	1.132
degraded_matches_severe_s0	degraded_matches_severe	success	failure	12	2	737	2.726
degraded_matches_severe_s1	degraded_matches_severe	success	failure	12	1	0	n/a
combined_severe_s0	combined_severe	success	failure	6	2	512	1.896
combined_severe_s1	combined_severe	success	failure	6	2	518	1.948

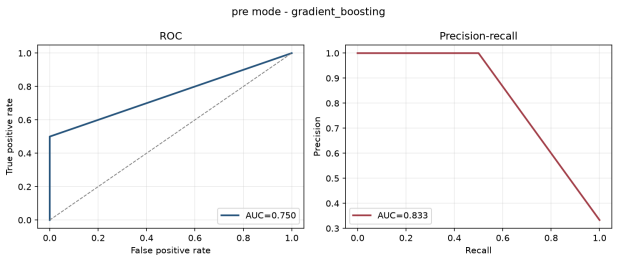
Synthetic failure model (pre)

Mode	pre
Leakage-free	True
Runs	16
Class counts	{'0': 9, '1': 7}
Features	38
Split	group_held_out
Best model	gradient_boosting
Threshold	0.5000
Preliminary	False

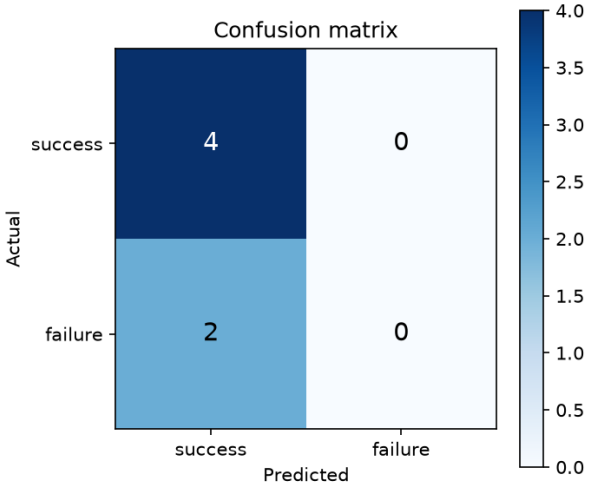
model	accuracy	precision	recall	f1	roc_auc	avg_prec	tp	tn	fp	fn
logistic_regression	0.667	0.000	0.000	0.000	0.625	0.500	0	4	0	2
random_forest	0.667	0.000	0.000	0.000	0.500	0.417	0	4	0	2
gradient_boosting	0.667	0.000	0.000	0.000	0.750	0.667	0	4	0	2

Held-out runs: blur_severe_s0, blur_severe_s1, clean_s0, clean_s1, limited_views_s0, limited_views_s1
Leave-one-condition-out (random forest):

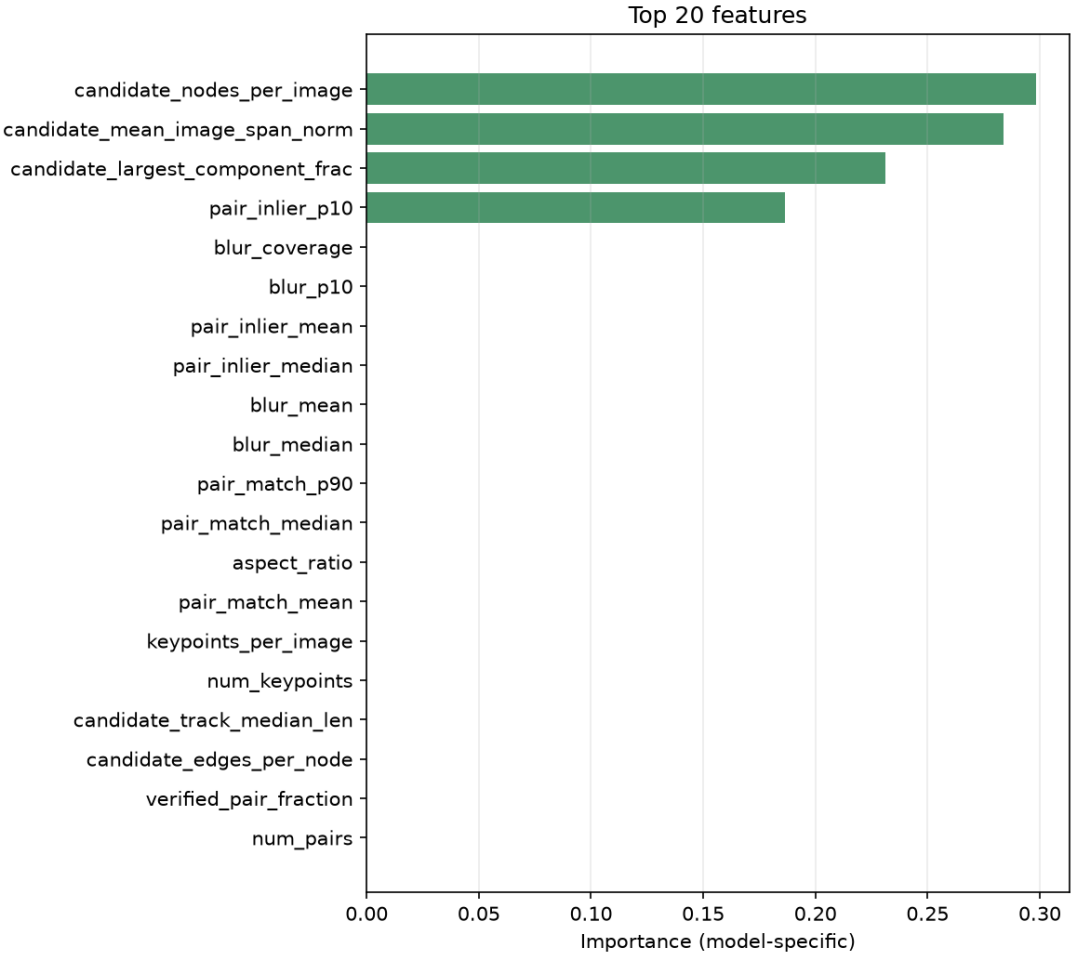
n	threshold	accuracy	precision	recall	f1	specificity	roc_auc	avg_prec	tp	tn	fp	fn	pred_pos	actual_pos
16.000	0.500	0.750	0.714	0.714	0.714	0.778	0.746	0.799	5.000	7.000	2.000	2.000	7.000	7.000



ROC / precision-recall



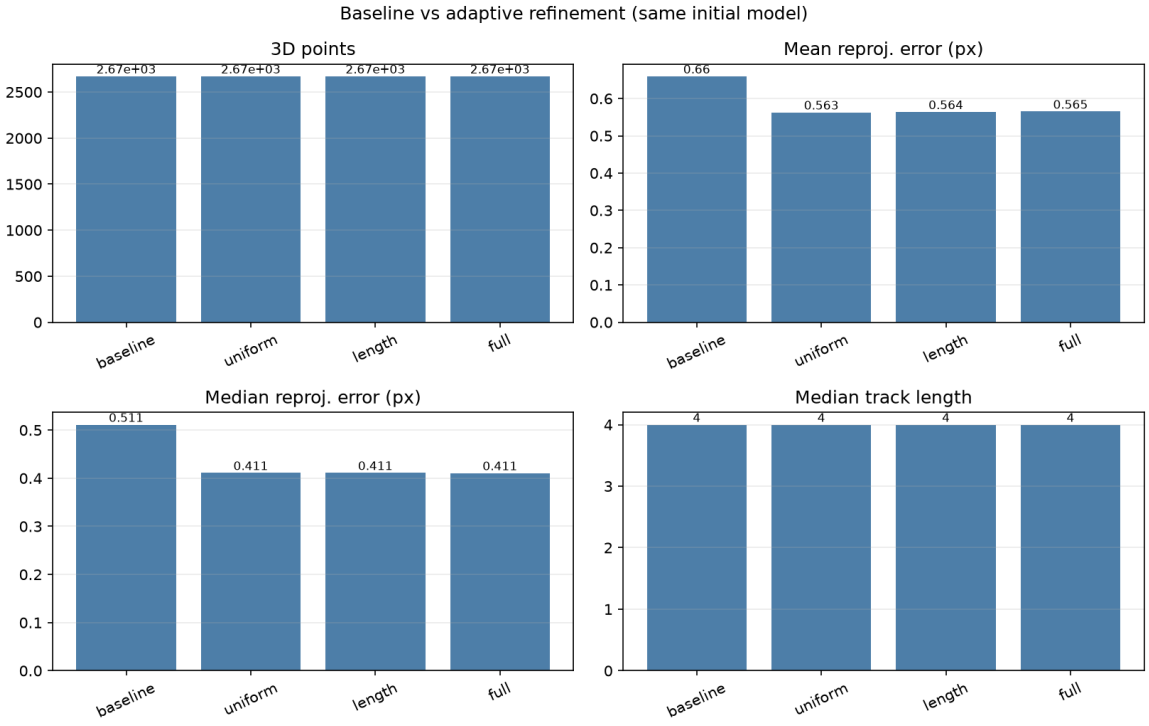
Confusion matrix



Feature importance

Synthetic adaptive comparison

variant	engine	points	mean_err	med_err	trk_med	success_label	sec
baseline	colmap_baseline	2672	0.660	0.511	4.000	0	2.147
uniform	scipy_least_squares	2672	0.563	0.411	4.000	0	9.900
length	scipy_least_squares	2672	0.564	0.411	4.000	0	8.533
full	scipy_least_squares	2672	0.565	0.411	4.000	0	8.596



Synthetic baseline vs refinement

Limitations

- Adaptive weighting is post-mapper refinement, not integrated into COLMAP's incremental mapper.
- scipy BA fixes intrinsics and gauge; it is not a Ceres/pyceres run.
- Failure labels and metrics are preliminary with few runs/conditions.
- Only ETH3D Courtyard and a synthetic scene are used; no cross-scene or ground-truth scan evaluation.
- pycolmap/pyceres were unavailable; COLMAP CLI conversion and scipy fallback were used.

Artifacts: outputs/runs, outputs/failure_model, outputs/experiments, outputs/REPORT.md, outputs/REPORT_SYNTHETIC.md